

Answer Key — Matter Is Made of Particles

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **constituent particles** — the basic building units are constituent particles.
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- Q2** (b) **breaks into particles that fit between water particles** — sugar splits into particles that occupy interparticle spaces.
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- Q3** (b) **physical change** — only the size changes; it stays chalk — physical change.
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- Q4** (b) **Acharya Kanad** — Acharya Kanad described Parmanu in the Vaisheshika Sutras.
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- Q5** (b) **False** — they are too small to see even with an ordinary microscope.
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- Q6** **interparticle** — the gaps are interparticle spaces.
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- Q7** **taste** — the sweet taste shows sugar is present.
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Short answer

- Q8** The smallest basic units that make up a substance, e.g. the tiny particles a piece of chalk or a grain of sugar is made of.
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- Q9** Sugar breaks into tiny constituent particles that spread into the interparticle spaces of water; they are too small to see, but their presence is sensed by the sweet taste.
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- Q10** No new substance is formed; the salt only separates into its particles and can be recovered, so it is a physical change.
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Long / application

- Q11** Grinding shows a chalk piece can be broken into smaller and smaller specks, each still chalk — suggesting it is made of many tiny particles. Dissolving sugar shows the sugar splits into particles too small to see that spread into the spaces between water particles (sensed by taste). Both show matter is made of large numbers of extremely small particles.
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- Q12** There are interparticle spaces between the water particles. The dissolved sugar particles are tiny and fit into these gaps, so very little extra volume is added and the water does not overflow much.
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